CarbonQuest – Carbon Footprint Tracking Platform

CarbonQuest is a cloud-based sustainability platform that helps users measure, track, and reduce their carbon footprint. The application focuses on awareness, data-driven insights, and simple actions that encourage environmentally responsible behavior.

The project demonstrates real-world application development with structured data handling, user authentication, and analytics-driven dashboards.

Features

Core Functionality

- User Authentication: Secure sign up and login system for personalized tracking
- Carbon Footprint Calculation: Calculate emissions based on daily activities such as travel, electricity usage, and consumption habits
- Activity Tracking: Log and manage carbon-generating activities over time
- Dashboard Analytics: Visual representation of emissions with charts and summaries
- Progress Tracking: Monitor emission trends and reduction progress
- Recommendations: Actionable tips to reduce carbon footprint

Sustainability Focus

- Emission calculations based on standard carbon estimation formulas
- Category-wise emission breakdown (transport, energy, lifestyle)
- Awareness-driven UI design to promote eco-friendly decisions

User Experience

- Simple and intuitive interface
- Responsive design for desktop and mobile
- Clean dashboards with clear data visualization
- Easy activity input and history tracking

Tech Stack

Frontend

- React
- Tailwind CSS

- Chart.js for data visualization

Backend & Cloud

- Node.js
- Express.js
- MongoDB for data storage
- JWT-based authentication

Database Schema

- Collections
- users – User profile and authentication data
- activities – Logged user activities related to carbon emissions
- emissions – Calculated emission records with timestamps

Getting Started

Prerequisites

- Node.js 18+
- MongoDB (local or cloud)

Setup Instructions

Clone and Install

```
git clone https://github.com/Vinayak-Kumar62/Carbon-Quest.git

cd Carbon-Quest

npm install
```

Configure Environment Variables

Create a .env file in the root directory:

```
PORT=5000

MONGO_URI=your_mongodb_connection_string

JWT_SECRET=your_jwt_secret
```

Run the Application

```
npm start
```

The application will start at:

```
http://localhost:5000
```

Usage Guide

Sign Up / Login

- Create a new account using email and password
- Log in to access your personalized dashboard

Add Activities

- Enter daily activities such as travel distance or electricity usage
- Submit data to calculate emissions instantly

View Dashboard

- Track total carbon footprint
- View category-wise emission breakdown
- Monitor progress over time

Improve Sustainability

- Follow suggested tips displayed on the dashboard
- Compare historical data to measure improvement

Architecture Highlights

Security

- JWT-based authentication
- Secure API routes
- User data isolation

Performance

- Optimized API calls
- Lightweight frontend components

- Efficient database queries

Scalability

- Modular backend architecture
- Easily extendable activity categories
- Cloud-ready deployment

API Design

User APIs

- Register and authenticate users
- Manage user sessions securely

Activity APIs

- Add, update, and delete activities
- Calculate emissions dynamically

Analytics APIs

- Fetch aggregated emission data
- Provide insights for dashboard charts

Interview Talking Points

What Makes This Project Stand Out

- Focuses on real-world sustainability challenges
- Combines analytics with environmental awareness
- End-to-end full-stack implementation
- Clear separation of frontend and backend logic

Technical Decisions

- MongoDB for flexible activity-based data
- JWT for stateless authentication
- React for dynamic UI updates
- Modular API design for future scalability

Project Structure

```
Carbon-Quest/
├── client/            # Frontend React app
│   ├── components/    # UI components
│   ├── pages/         # Application pages
│   └── utils/         # Helper functions
├── server/            # Backend Node.js app
│   ├── routes/        # API routes
│   ├── models/        # Database schemas
│   ├── controllers/   # Business logic
│   └── middleware/    # Auth and validation
└── README.md
```

Future Enhancements

- Carbon offset integration
- Goal-based emission reduction tracking
- Export reports as PDF
- Admin dashboard for analytics
- Mobile application
- Community challenges and leaderboards

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